Empirical rule

1 In a normal distribution, state the approximate percentage of the scores that lie within:

a 1 standard deviation of the mean.

b 2 standard deviations of the mean.

c 3 standard deviations of the mean.

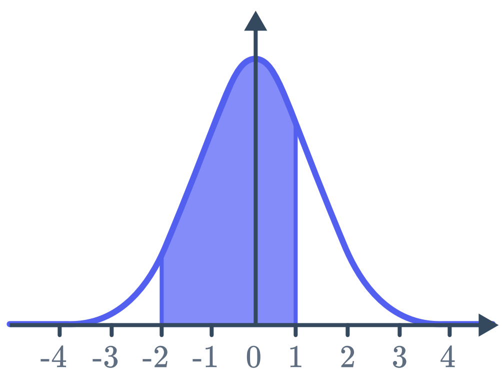
2 In a normal distribution, state the approximate percentage of scores that lie between the mean and:

a 1 standard deviation above the mean.

b 2 standard deviations below the mean.

3 In the following normal distributions, each unit on the horizontal axis indicates 1 standard deviation. Find the approximate percentage of scores that lie in the shaded region:

a



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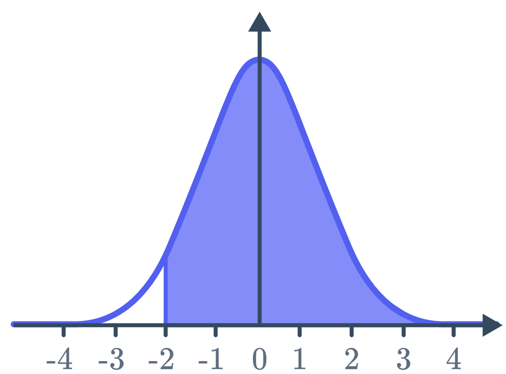
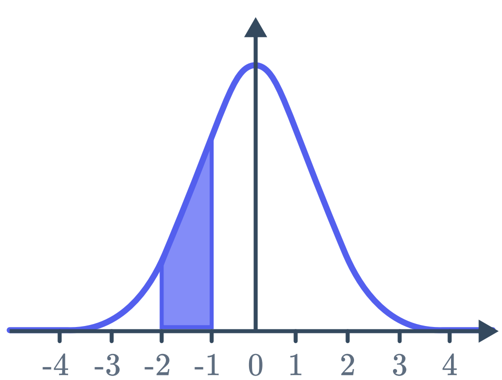


c



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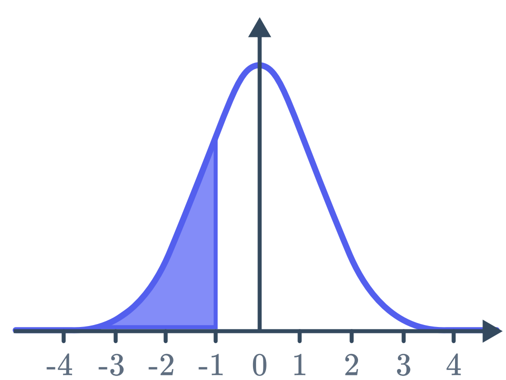

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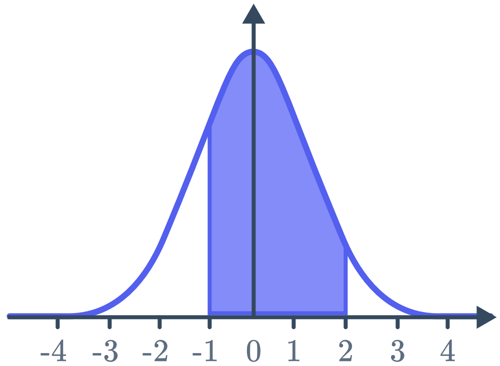

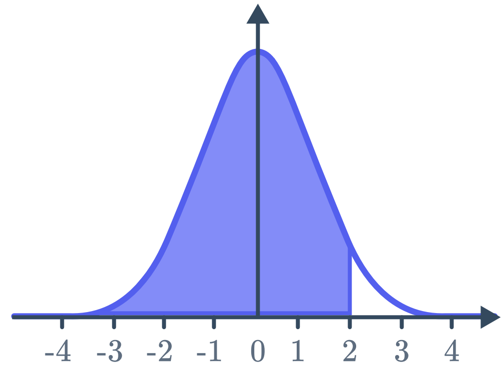

4 In a normal distribution, state the approximate percentage of scores that lie between:

a 1 standard deviation above and 2 standard deviations below the mean.

b 1 standard deviation below and 3 standard deviations above the mean.

c 2 standard deviations below and 3 standard deviations above the mean.

5 A set of scores is approximately normally distributed, where the mean score is 92 and standard deviation is 20. Using the empirical rule, find the percentage of scores between:

a 72 and 112.
b 52 and 132.
c 32 and 152.
d 92 and 112.
e 112 and 132.

6 The exams scores of students are approximately normally distributed with a mean score of 63 and a standard deviation of 8. Use the empirical rule to find:

a The approximate percentage of students who scored between 39 and 79.

b The approximate number of students who passed if there are 450 students in the class and if the passing score is 39.

7 For a set of scores, it was found that the mean score was 61 and the standard deviation was 11. If the distribution of scores is approximately normal, find the percentage of scores between:

a 50 and 72.
b 39 and 83.
c 28 and 94.
d 61 and 72.
e 72 and 83.

8 The mean of a set of scores is 77 and the standard deviation is 29. Find the value of:

a $\text{Mean } - \text{Standard Deviation}$

b $\text{Mean } + 2 \times \text{Standard Deviation}$

c $\text{Mean } - \dfrac{(2 \times \text{Standard Deviation})}{3}$

d $\text{Mean } + \dfrac{(4 \times \text{Standard Deviation})}{5}$

e $\text{Mean } + 3 \times \text{Standard Deviation}$

f $\text{Mean } - 2 \times \text{Standard Deviation}$

9 The mean of a set of scores, denoted by μ is 51 and the standard deviation, denoted by σ is 16. Find the value of the following:

a $\mu - \sigma$

b $\mu + 2\sigma$

c $\mu + 3\sigma$

d $\mu - 2\sigma$

e $\mu + 0.5\sigma$

f $\mu - \dfrac{2\sigma}{3}$

10 The heights of 500 boys are found to approximately follow such a distribution, with a mean height of 145 cm and a standard deviation of 11 cm. Find the number of boys with heights between:

a 134 cm and 156 cm.

b 112 cm and 178 cm.

11 The times that a class of students spent talking or texting on their phones on a particular weekend is approximately normally distributed with mean time 173 minutes and standard deviation 4 minutes. Find the percentage of students that used their phones for between 165 and 181 minutes on the weekend.

12 The operating times of phone batteries are approximately normally distributed with mean 34 hours and a standard deviation of 4 hours.

a Find the percentage of batteries that last between 22 and 42 hours.

b Find the percentage of batteries that last between 30 hours and 42 hours.

c Any battery that lasts less than 22 hours is deemed faulty. If a company manufactured $51\,000$ batteries, find the approximate number of batteries they would be able to sell.

13 The number of biscuits in a box is approximately normally distributed with mean 30 and standard deviation of 3.

a Approximately 81.5% of the scores lie between 2 standard deviations below and x standard deviation(s) above the mean. Find the value of x.

b Find the range of the numbers of biscuits in 81.5% of the boxes.

14 The times for runners to complete a 100 m race is approximately normally distributed with mean 14 seconds and standard deviation of 1.9 seconds.
Find the range of time in seconds for which 97.35% of runners completed the race. Round your answer to one decimal place.

15 The number of biscuits packaged in biscuit boxes is approximately normally distributed with mean 38 and standard deviation 5. If 4000 boxes of biscuits are produced, find the approximate number of boxes with more than 33 biscuits.

16 The times that professional divers can hold their breath are approximately normally distributed with mean 106 seconds and standard deviation 8 seconds. If 700 professional divers are selected at random, find the approximate number of divers that would be able to hold their breath for longer than 82 seconds.

17 The literacy rate of a population is used to help measure the level of development of a country. The mean literacy rate in a particular country is 59%, and the standard deviation is 5%. The literacy rate varies from place to place within the country.

a Find the literacy rate that is 3 standard deviations above the mean. Write your answer as a percentage.

b Find the literacy rate that is 2 standard deviations below the mean. Write your answer as a percentage.

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z-scores

18 Dylan scored 90% with a z-score of 2 in English, and 78% with a z-score of 4 in Mathematics. In which subject was his performance better, relative to the rest of his class?

19 Jenny scored 81% with a z- score of -2 in English, and 72% with a z-score of -4 in Mathematics. In which subject was her performance better, relative to the rest of the class?

20 If Dave scores 96 in a test that has a mean score of 128 and a standard deviation of 16, what is his z-score?

21 If Harrison scores 57.6, with a z-score of 2, in a test that has a standard deviation of 5.8, what was the mean score?

22 Ivan scored 55.25 in his test, in which the mean was 64.5 and the standard deviation was 2.5. Maria scored 50.22 in her test, in which the mean was 57.9 and the standard deviation was 2.4.

a Find Ivan's z-score.

b Find Maria's z-score.

c Whose performance was better relative to the other students in their classes?

23 If Luke scores 68, for a z-score of -3, in a test that has a mean score of 93.5, what was the standard deviation of the test scores?

24 A particular investment fund has returned 17.2% p.a. on average over a period. If the mean return of all investment funds over the same period was 8% p.a. and the standard deviation was 2.3%, what is this fund's z-score?

25 A general ability test has a mean score of 100 and a standard deviation of 15.

a If Paul received a score of 102 in the test, what was his z-score correct to two decimal places?

b If Georgia had a z-score of 3.13, what was her score in the test, correct to the nearest integer?

26 For each of the following examples, find x, the test score each student received:

a Iain's z-score in a test is 1, the mean score is 62% and standard deviation is 3%.

b Rochelle's z-score in a test is -3, the mean score is 75% and standard deviation is 3%.

c Aaron's z-score in a test is 3.6, the mean score is 75% and standard deviation is 3%.

d Sean's z-score in a test is -1.1, the mean score is 76% and standard deviation is 3%.

27 Ray scored 12.49 in his test, in which the mean was 7.9 and the standard deviation was 1.7. Gwen scored 30.56 in her test, in which the mean was 20.2 and the standard deviation was 2.8.

a Find Ray's z-score.

b Find Gwen's z-score.

c Which of the two had a better performance relative to the other students in their classes?

28 Marge scored 43 in her Mathematics exam, in which the mean score was 49 and the standard deviation was 5. She also scored 92.2 in her Philosophy exam, in which the mean score was 98 and the standard deviation was 2.

a Find Marge's z-score in Mathematics.

b Find Marge's z-score in Philosophy.

c Which exam did Marge do better in, compared to the rest of her class?

29 Kathleen scored 83.4 in her Biology exam, in which the mean score and standard deviation were 81 and 2 respectively. She also scored 60 in her Geography exam, in which the mean score was 46 and the standard deviation was 4.

a Find Kathleen's z-score in Biology. Round your answer to one decimal place.

b Find Kathleen's z-score in Geography. Round your answer to one decimal place.

c Which exam did Kathleen do better in?

30 Frank finishes a fun run in 156 minutes. If the mean time taken to finish the race is 120 minutes and the standard deviation is 12 minutes, what is his z-score?

31 Christa is 157 cm tall. If the average height in her class is 141 cm and the standard deviation is 8 cm, what is her z-score?

32 The number of runs scored by Tobias in each of his innings is listed below:

$$33,\ 31,\ 32,\ 30,\ 32,\ 30,\ 32,\ 30,\ 31,\ 34$$

Find each of the following, rounding your answers to two decimal places:

a His batting average.

b His population standard deviation.

c The z-score of his final innings.

d The z-score of his greatest scoring innings.

33 Packets of flour are each labeled as having [redacted]s of 1 kg. The mass of these packets is normally distributed with a mean of 1.06 kg and a standard deviation of 0.03 kg.

a Complete the following table:

Mass (kg)	1	1.03	1.06	1.09	1.12
z-score					

b What percentage of packets will have a mass less than 1.06 kg?

34 Buzz’s best time in the half marathon is 90.2 minutes, while his best time in the full marathon is 185.6 minutes.

a The mean time to complete the half marathon is 110 minutes with a standard deviation of 6 minutes among world class runners. What is the z-score of Buzz’s best time in the half marathon?

b The mean time to complete the full marathon is 200 minutes with a standard deviation of 3 minutes among world class runners. What is the z-score of Buzz’s best time in the full marathon?

c In which event does Buzz have a better 'best time', relative to world class runners?

35 A machine is set for the production of cylinders of mean diameter 5.06 in, with standard deviation 0.016 in.

a Assuming a normal distribution, between what values, in inches, will 99.7% of the diameters lie?

b If cylinders with diameters less than 5.012 in or more than 5.108 in are discarded, what percentage of cylinders produced are discarded?

c If a cylinder, randomly selected from this production, has a diameter of 5.124 in, what conclusion could be drawn?

36 Athletes take a lesser heart rate as a sign of better fitness. Victoria and John record their heart rates in beats per minute (bpm) each time they finish a triathlon. Both their heart rates are normally distributed, with Victoria’s $\mu = 170$ beats, $\sigma = 9$ beats and John's $\mu = 161$ beats, $\sigma = 8$ beats. At the end of their most recent race, Victoria’s heart rate was 192 bpm and John’s was 172 bpm.

a Calculate Victoria’s z-score in the most recent race to one decimal place.

b Calculate John’s z-score in the most recent race to one decimal place.

c Who’s heart rate was better in the most recent race?

37 The table records the lengths of fish a fisherman caught on a particular day:

Type	Length (cm)
Yellowfin Bream	$28, 26, 28, 26, 29$
Flathead	$34, 36, 33, 35, 32, 36$

a Calculate the mean length of the Yellowfish Bream correct to two decimal places.

b Calculate the sample standard deviation of the length of a Yellowfin bream correct to two decimal places.

c The next Yellowfin bream he catches is 30.40 cm , calculate the z-score of this fish correct to one decimal place.

d Calculate the mean length of the Flathead correct to two decimal places.

e Calculate the standard deviation of the length of a Flathead correct to two decimal places.

f The next Flathead he catches is 37.33 cm, calculate the z-score of this fish correct to one decimal place.

g Which of these two fish is longer, relative to their cohort?